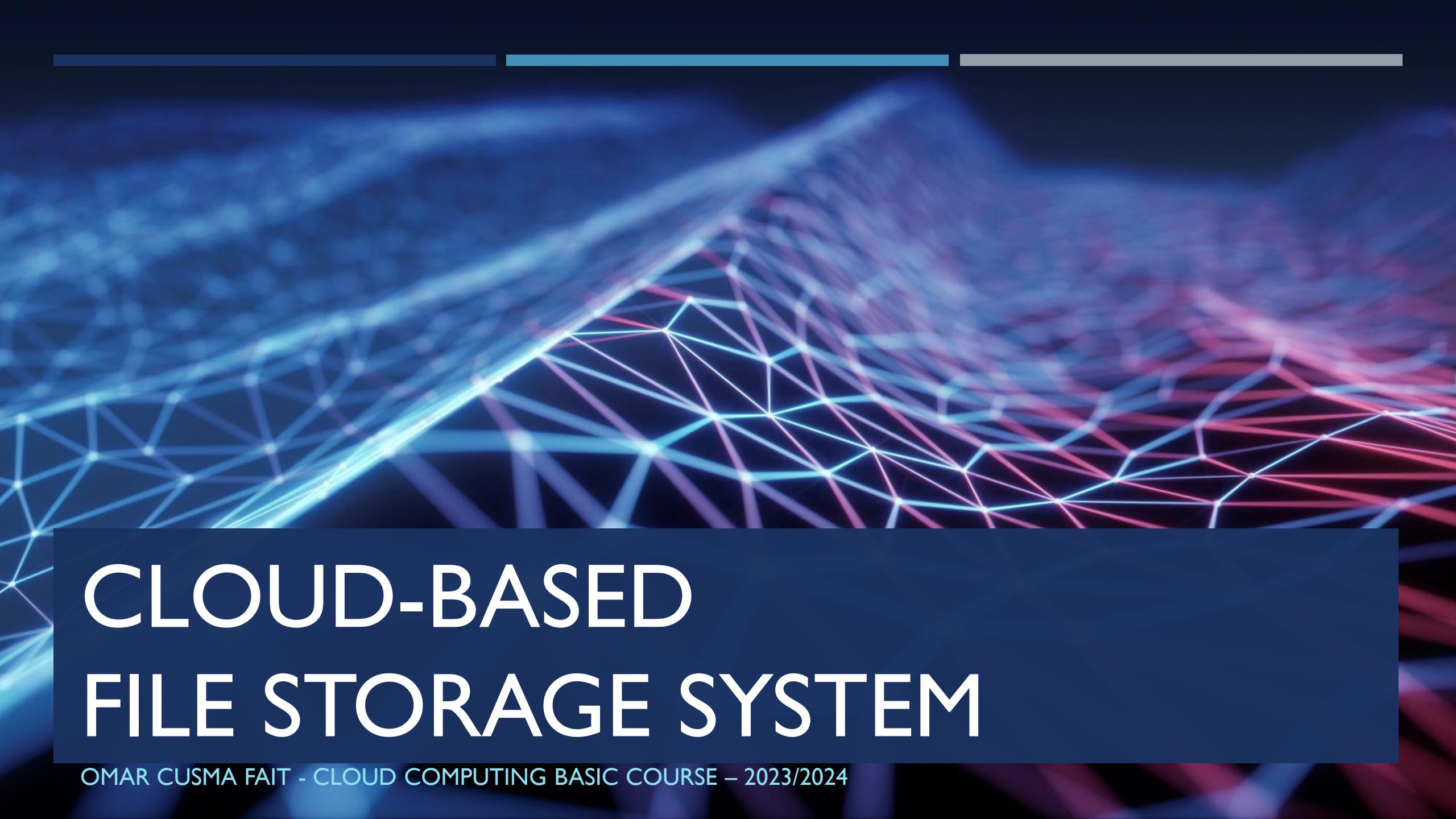




CLOUD-BASED FILE STORAGE SYSTEM

OMAR CUSMA FAIT - CLOUD COMPUTING BASIC COURSE – 2023/2024

PROJECT OVERVIEW

Objective:

Deploy a self-hosted cloud storage platform with monitoring and performance testing capabilities



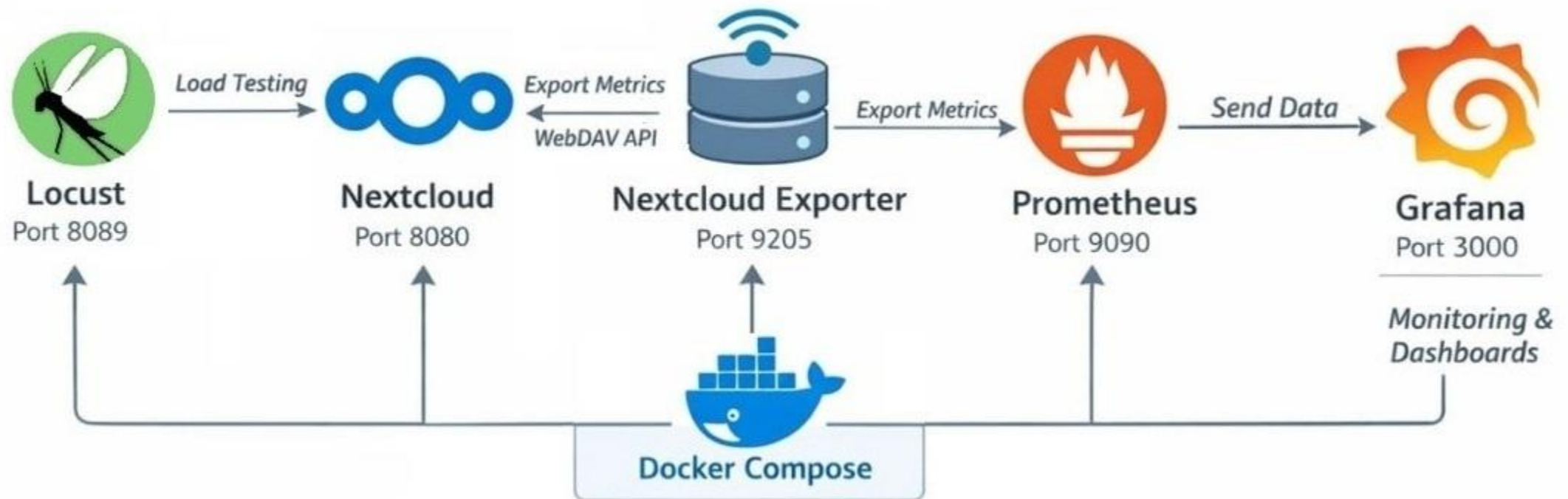
Requirements Addressed:

- ✓ User authentication & authorization
- ✓ Private storage spaces
- ✓ File operations (upload/download/delete)
- ✓ Scalability, security, and cost-efficiency

Key Components:

- **Nextcloud:** File storage and user management
- **Docker Compose:** Container orchestration
- **Prometheus + Grafana:** Monitoring and visualization
- **Locust:** Load testing and performance analysis

SYSTEM ARCHITECTURE



EXTCLOUD FEATURES



Accedi a Nextcloud

Accedi con nome utente o email

Password



→ Accedi

Hai dimenticato la password?

Accedi con un dispositivo

Admin Capabilities:

- ✓ User account creation and management
- ✓ Private storage allocation per user (`/remote.php/dav/files/username/`)
- ✓ Group management and permissions
- ✓ System configuration and security settings

User Capabilities:

- ✓ Upload/download/delete files
- ✓ Sign-up, log out
- ✓ File preview (images, PDFs, text)
- ✓ Folder creation and organization
- ✓ Web interface and desktop sync

MONITORING STACK



Nextcloud Exporter

- Scrapes Nextcloud internal metrics
- Exposes data at ``http://localhost:9205/metrics``
- Key metrics: active users, file count, response times



Prometheus

- Time-series database
- Scrape interval: 30 seconds
- Configuration: ``prometheus/prometheus.yml``



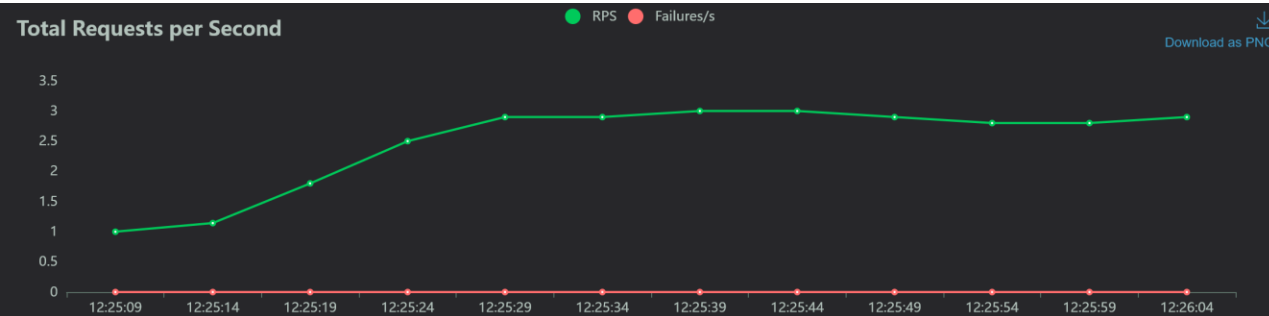
Grafana

- Auto-provisioned Prometheus datasource
- Custom dashboards for real-time monitoring
- Access: ``http://localhost:3000``

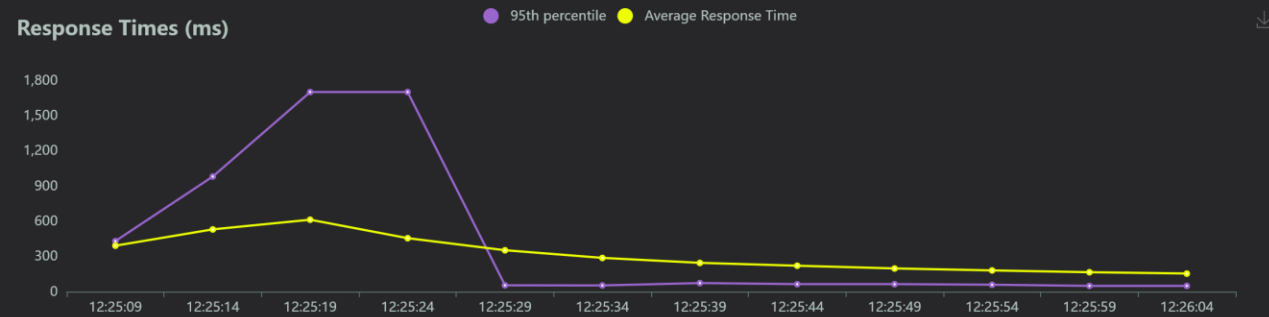
LOAD-TESTING RESULTS: 500KB - 5MB



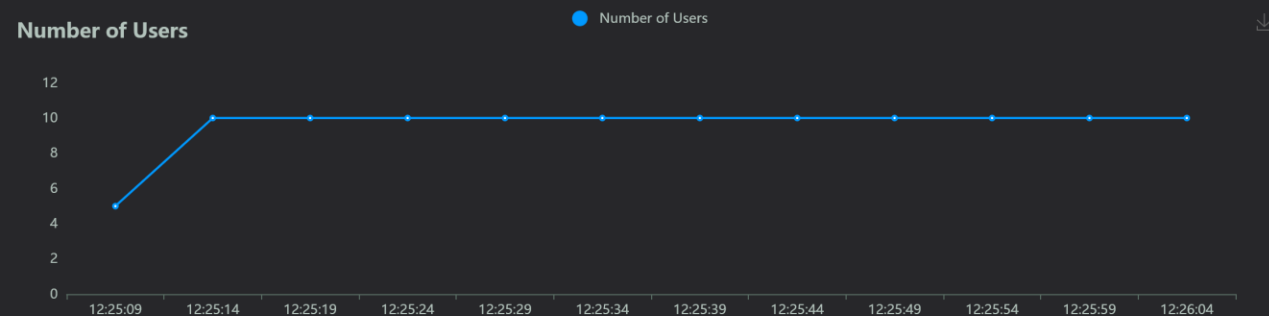
Total Requests per Second



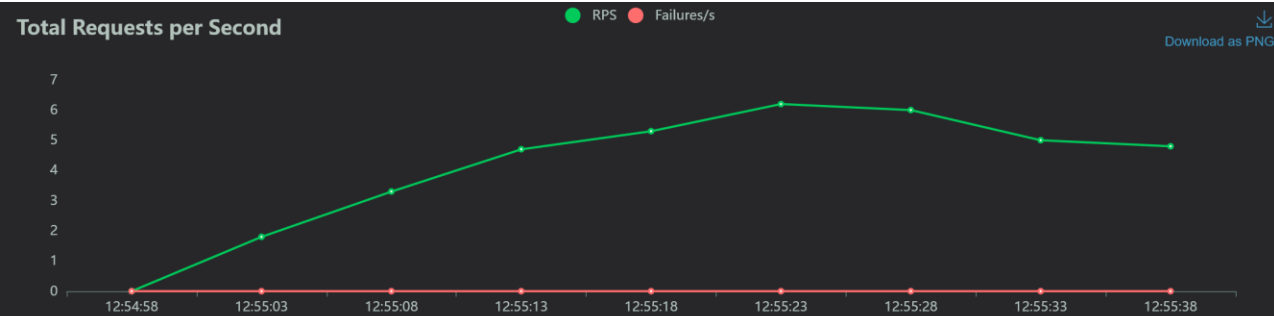
Response Times (ms)



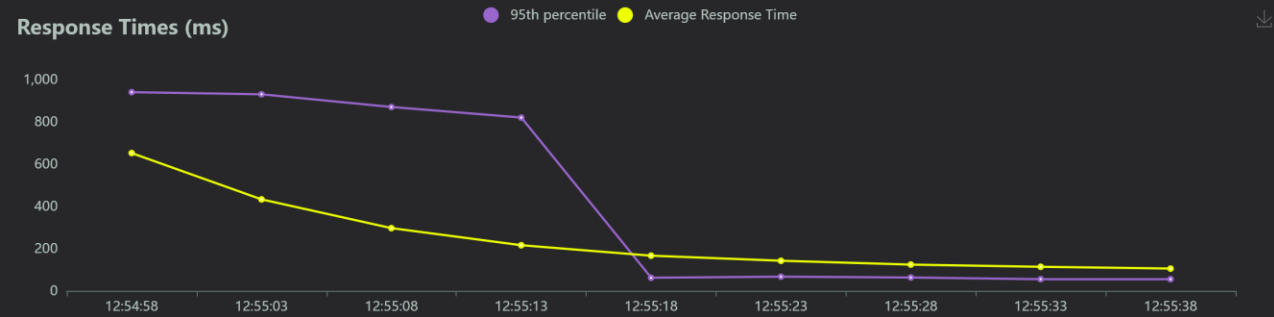
Number of Users



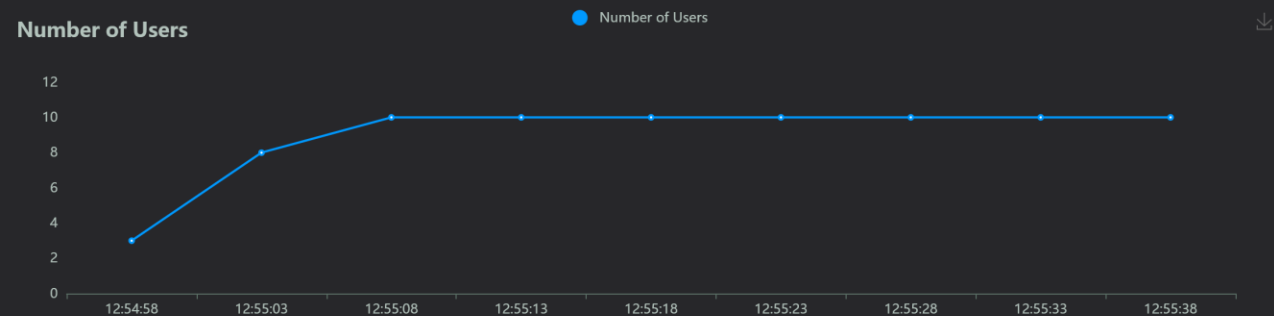
Total Requests per Second



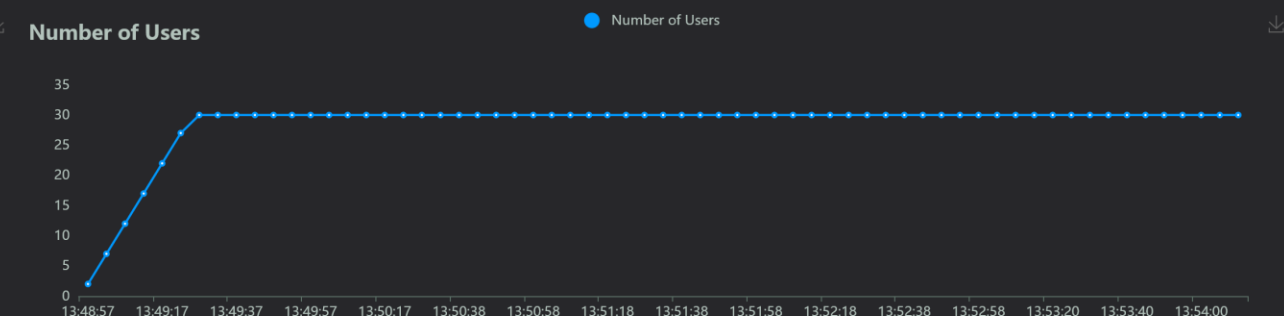
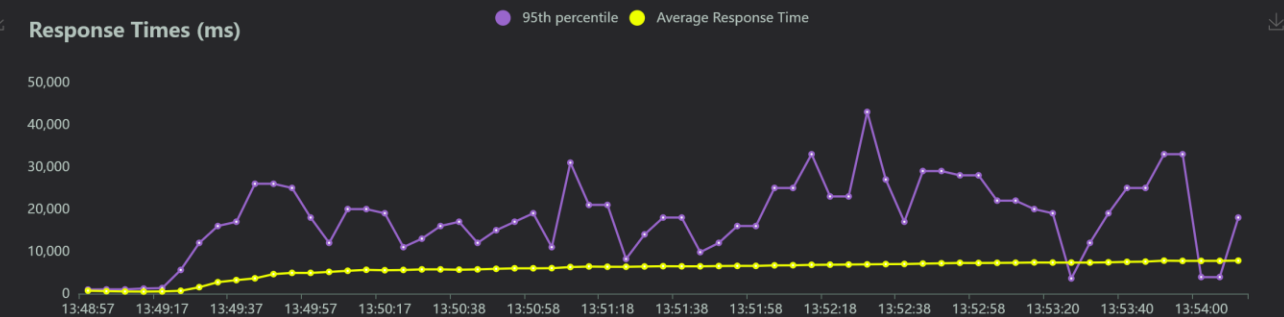
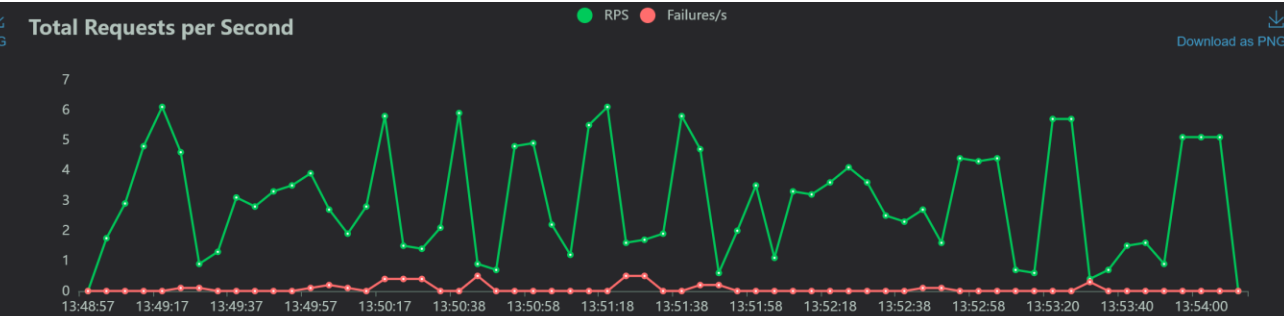
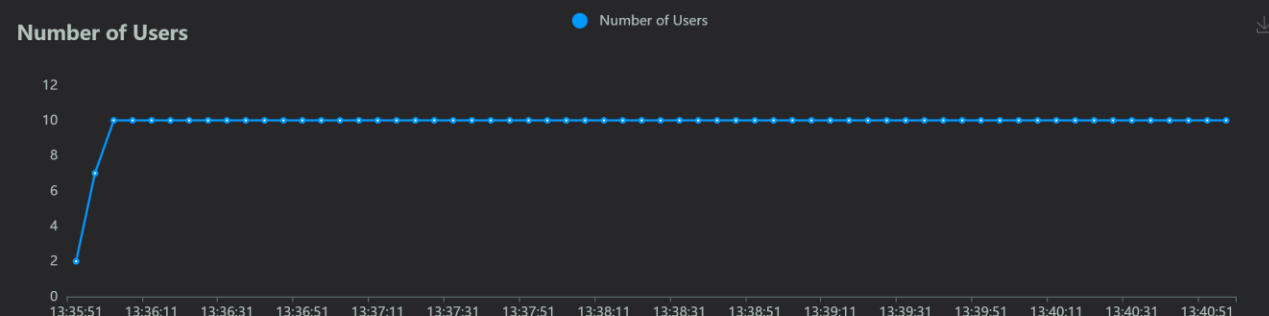
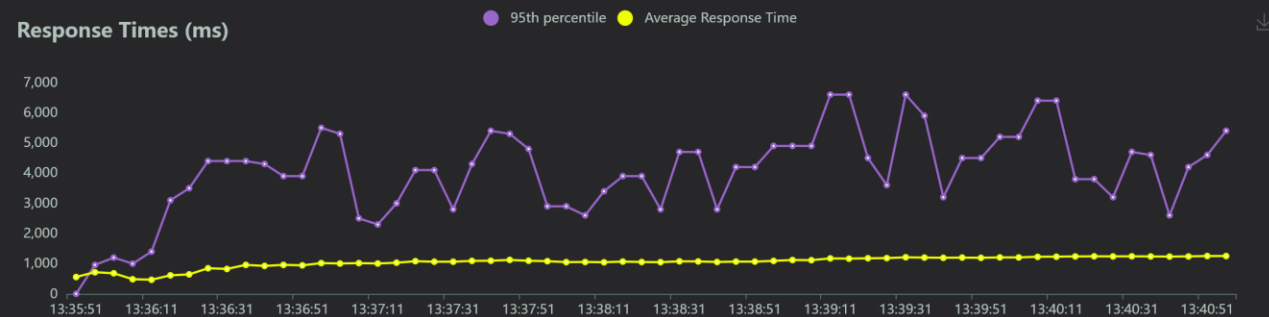
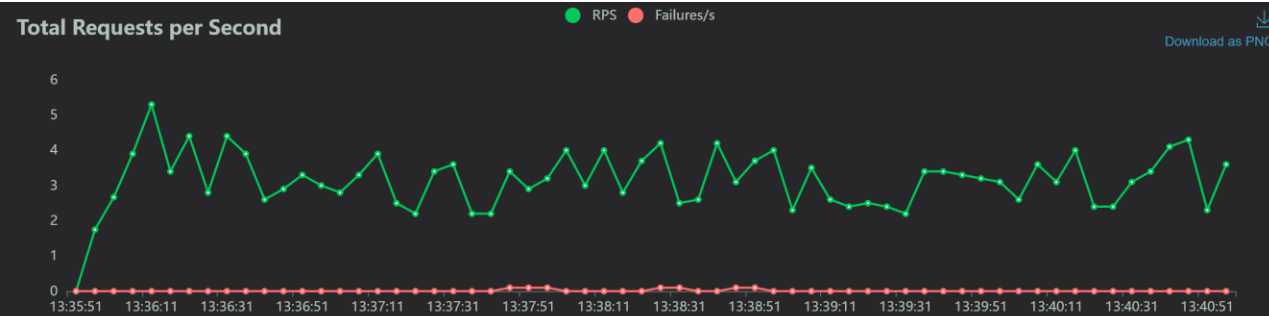
Response Times (ms)



Number of Users

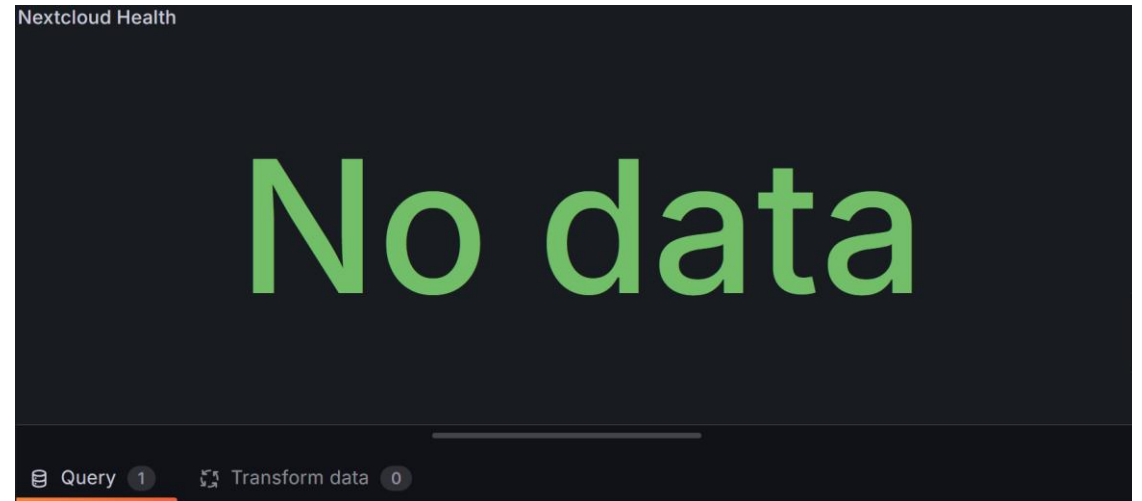


LOAD-TESTING RESULTS: 50MB



CHALLENGES & SOLUTIONS

- The **Nextcloud Exporter** was unable to pull metrics from Nextcloud.
 - Nextcloud was blocking the exporter because it was scraping too frequently.
 - This resulted in: `nextcloud_up = 0`
- WSL files from the **stress-test** almost **filled up the disk** with thousands of 50MB temporary files
 - deleting them didn't help!
 - The Windows diskpart tool saved the day
- *Many small issues... but I didn't give up!*

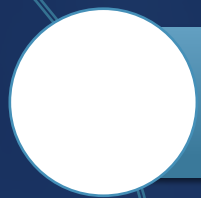


```
C:\Users>quit
"quit" non è riconosciuto come comando interno o esterno,
un programma eseguibile o un file batch.
```

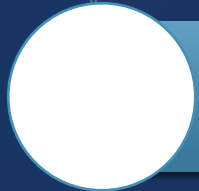

ACHIEVEMENTS

- ✓ Fully containerized Nextcloud deployment
- ✓ Complete monitoring stack (Prometheus + Grafana)
- ✓ Automated load testing with Locust
- ✓ Comprehensive documentation and execution logs

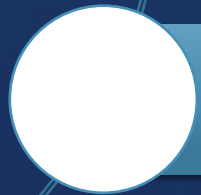
PERFORMANCE INSIGHTS



Medium files (5MB):
median response ~44–49 ms at 10 concurrent users



System **handles well** moderate concurrent load



Load testing shows that performance is consistent for
any small number of users

SCALING AND OTHER IMPROVEMENTS



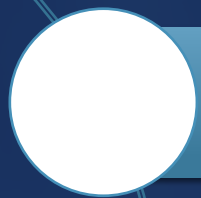
Scalability: PostgreSQL + Redis backend; horizontal scaling

Security: Add TLS/SSL via reverse proxy (Traefik/Nginx)

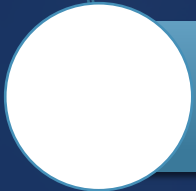
Cloud Deployment: AWS EC2 + S3 with auto-scaling

Larger Scale Tests: GB-sized files and 100+ concurrent users

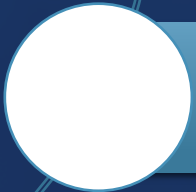
COST-EFFICIENCY



All **open-source** components (zero licensing)



Lightweight containers optimize resource usage



Cloud deployment would use spot instances for cost optimization



THANK YOU